TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Cloth Thickness

ISO 5084 in practice: why a fabric has no single thickness, how pressure chooses the answer, and what compressibility says about handle and bulk.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 3 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the thickness of the reference fabric by the ISO 5084 dead-weight gauge principle at a reference pressure, to measure thickness at low and high pressures and hence its percentage compression, and to relate thickness to mass per unit area.

1.1 The ISO 5084 principle

ISO 5084 defines thickness as the distance between a reference plate (anvil) and a parallel circular presser foot resting on the fabric, under a specified pressure applied for a specified time. Three quantities therefore define every thickness result: **presser-foot area**, **applied pressure** and **dwell time**. A widely used combination is a foot of 20 cm² (diameter about 50.5 mm) at a pressure of 1 kPa, with the reading taken after the foot has rested on the fabric for a standard dwell (commonly 30 s) so that rapid initial compression has finished. The gauge itself — dead-weight, spring or electronic — is immaterial provided it resolves 0.01 mm and applies the stated pressure without impact.

$$\text{Pressure (kPa)} = \text{applied force (N)} / \text{presser-foot area (m}^2\text{)}$$

1.2 The pressure–thickness relation

A fabric is a compressible assembly of yarns and trapped air. At very low pressure the foot rests on surface hairs and the crowns of the yarns — the fabric reads thick. As pressure rises, hairs flatten, yarns compress and inter-yarn air is expelled, and the thickness falls steeply at first, then more slowly as the yarn cores themselves resist. The thickness–pressure curve is therefore steep at low pressure and flattens at high pressure: *the choice of pressure chooses the result*. Two laboratories using different pressures will report different thicknesses for the identical fabric, both correctly. This is why specifications must state the pressure, and why Experiment 1 quoted its survey value of 0.434 mm together with its reference pressure.

1.3 Compressibility

The fall in thickness between two pressures is itself a useful property — it measures bulk and softness of handle:

$$\% \text{ compression} = (t_{low} - t_{high}) / t_{low} \times 100$$

A lofty, softly finished fabric may compress 30–40% between light and firm pressure; a dense, calendered fabric compresses only a few percent. Compressibility correlates with the fullness of handle, seam behaviour and the thickness a garment actually occupies in a packed carton — all governed by the low-pressure, uncompressed bulk.

1.4 Thickness and GSM

Thickness and GSM (Experiment 2) together give the fabric's apparent density:

$$\text{Apparent density} = \text{GSM} / \text{thickness} = 185.5 \text{ g/m}^2 / 0.434 \text{ mm} \approx 0.427 \text{ g/cm}^3$$

The reference fabric packs 185.5 g into each square metre at 0.434 mm, an apparent density of about 0.43 g/cm³ — well below solid cotton (about 1.5 g/cm³), because roughly two-thirds of the fabric volume is air (see also the porosity of Experiment 5). Two fabrics of equal GSM but different thickness differ exactly in this trapped air: the thicker one is loftier, warmer and softer; the thinner one denser, crisper and usually stronger per unit weight.

All testing is at the standard atmosphere, 65% RH and 20 °C, on conditioned fabric.

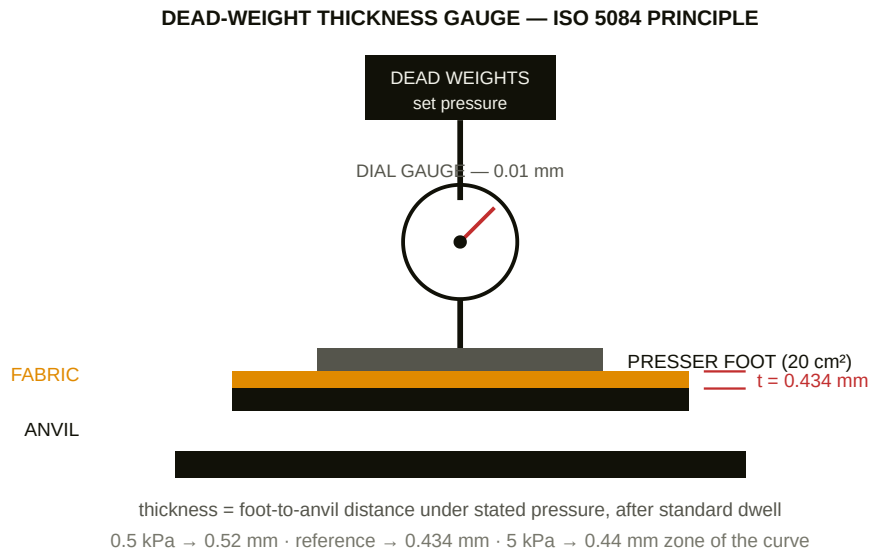


Figure 1 — Dead-weight thickness gauge: interchangeable weights set the pressure on a circular presser foot, the dial reads the foot-to-anvil distance through the fabric, and the value belongs to the stated pressure and dwell time.

2. Apparatus and procedure

- **Apparatus:** dead-weight thickness gauge with interchangeable presser feet and weights, dial or digital readout to 0.01 mm (ISO 5084 arrangement); conditioned reference fabric; stopwatch for dwell timing.
- **Zero check:** with no fabric, lower the foot onto the anvil under the test pressure and confirm zero. Clean both faces — a single loose fibre under the foot is a false 0.02 mm.
- **Reference-pressure measurement:** place the fabric flat and unwrinkled between foot and anvil at a fresh position, lower the foot gently (never drop it), allow the standard dwell, and read to 0.01 mm. Take ten readings at spread positions, avoiding selvedges, creases and seams, and average.
- **Two-pressure measurement:** repeat the measurement at a low pressure of 0.5 kPa and at a high pressure of 5 kPa, using the appropriate weights and the same positions, allowing the same dwell at each pressure.
- **Calculation:** report the reference-pressure mean, then compute percentage compression between the 0.5 kPa and 5 kPa thicknesses and the apparent density from GSM and reference thickness.

3. Worked calculation

Reference pressure. Ten readings lay between 0.42 mm and 0.45 mm with mean:

$$\text{Mean thickness at reference pressure} = 0.434 \text{ mm}$$

This is the value carried from Experiment 1 — the thickness of record for the 70 × 59, 185.5 g/m² reference fabric.

Effect of pressure. The same positions measured at two further pressures gave:

PRESSURE	MEAN THICKNESS	READING CHARACTER
0.5 kPa (low)	0.52 mm	Foot rests on yarn crowns and surface hair — full bulk
Reference pressure	0.434 mm	Thickness of record
5 kPa (high)	0.44 mm	Surface flattened, yarns partly compressed

$$\% \text{ compression} = (0.52 - 0.44) / 0.52 \times 100 = 0.08 / 0.52 \times 100 = 15.4\%$$

Apparent density.

$$\text{Apparent density} = 185.5 \text{ g/m}^2 / 0.000434 \text{ m} = 427 \text{ kg/m}^3 = 0.427 \text{ g/cm}^3$$

> **READING THE RESULT**
>
> One fabric, three honest thicknesses: 0.52 mm at 0.5 kPa, 0.434 mm at the reference pressure and 0.44 mm at 5 kPa, with 15.4% compression between the extremes — a moderately firm fabric,

> neither lofty nor boardy. Any quotation of thickness without pressure is incomplete; the compression percentage, not any single reading, is the handle information.

4. Precautions

- Lower the presser foot gently every time; impact loading momentarily over-compresses and the reading creeps back during dwell, scattering the set.
- Respect the dwell time at every reading — thickness falls continuously at first, so readings taken at random dwells are not comparable.
- Measure at fresh, spread positions; re-measuring one spot flattens it progressively (the fabric remembers the foot).
- Avoid selvedges, creases, seams and printed or embossed areas, all of which are locally thicker than the ground fabric.
- Change weights, not foot area, when changing pressure where possible — pressure and foot size are separate variables in the standard.
- Condition at 65% RH, 20 °C: a swollen humid fabric is measurably thicker, and compression behaviour changes with regain.

Practice (viva preparation)

1. State the ISO 5084 definition of fabric thickness, naming the three quantities that must accompany every result.
2. Thickness is 0.52 mm at 0.5 kPa and 0.44 mm at 5 kPa. Find the percentage compression.
3. Why do two laboratories report different thicknesses for the same fabric, both correctly?
4. Find the apparent density of a fabric of 185.5 g/m² and 0.434 mm thickness, and interpret the value against solid cotton at about 1.5 g/cm³.
5. Why is the presser foot lowered gently and the reading delayed by a standard dwell?
6. Two fabrics have the same GSM but thicknesses of 0.35 mm and 0.55 mm. How will they differ in use?
7. Why must thickness positions avoid selvedges and creases?

Answer key

1. Thickness is the distance between the anvil (reference plate) and the parallel circular presser foot resting on the fabric. Every result must be accompanied by the presser-foot area, the applied pressure and the dwell time before reading — ISO 5084 standardises all three together with the 0.01 mm resolution.

2. % compression = (0.52 − 0.44) / 0.52 × 100 = 15.4%.
3. Because thickness is pressure-dependent: the fabric compresses as pressure rises, steeply at first then more slowly. Different test pressures place the foot at different points on the same thickness–pressure curve. Both results describe the fabric correctly — at their own stated pressures — which is why a pressure must always be quoted.
4. Apparent density = GSM / thickness = 185.5 / 0.434 ≈ 0.427 g/cm³ (427 kg/m³). It is only about 28% of the density of solid cotton fibre, so roughly 70% of the fabric's volume is trapped air — consistent with the open-area porosity found in Experiment 5 and the source of the fabric's insulation and softness.
5. A dropped foot adds impact compression that recovers only slowly, and even a gently lowered foot continues to compress the yarn assembly for some seconds as air escapes and fibres rearrange. The standard dwell lets that rapid phase finish so every reading is taken at the same point of the compression–time curve.
6. Equal mass spread over different bulk: the 0.55 mm fabric holds more trapped air — it will feel fuller and softer, insulate better and occupy more carton volume; the 0.35 mm fabric is denser and crisper, drapes flatter and typically gives more strength and abrasion life per unit weight. GSM alone cannot distinguish them; thickness completes the picture.
7. Selvedges are woven denser and often with heavier yarns, and creases double the fabric locally. Both read thicker than the ground cloth and would inflate the mean and its scatter; the specification concerns the body of the fabric, so readings are taken on flat ground areas at least a few centimetres from any edge or fold.

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